Formalizing the Local-Global Gap in Symbolic World Models: The Causal Decoupling Theorem and Topological Divergence Bounds

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Abstract

Symbolic action models (STRIPS/PDDL) provide the foundation for model-based planning, heuristic search, and counterfactual simulation. A pervasive assumption in Action Model Learning (AML) and program-synthesized world models is that high passive transition prediction accuracy (A_{pred}) over observed traces guarantees reliable downstream plan execution. In this work, we prove that this assumption is fundamentally flawed in combinatorial search graphs. We formalize the *Search-Tree Topology Collapse* phenomenon under sparse precondition omissions. We establish: (i) the *No-Edges-Lost Lemma* (Theorem 7), proving that precondition omissions strictly introduce spurious phantom edges; (ii) the *Search-Tree Phantom Path Divergence Proposition* (Theorem 8), proving an exponential divergence $d_{\triangle} = \Omega(b^{D-d})$ while passive accuracy satisfies $A_{\text{pred}} \geq 1 - b^{-d}$; and (iii) the *Causal Decoupling Proposition* (Theorem 9), proving that A_{pred} and Plan Execution Success Rate (PESR) are statistically and functionally independent. These results demonstrate why passive trajectory evaluation is incapable of certifying world models for goal-directed autonomous agents.

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1 Introduction and Foundational Notations

Let $M^* = \langle \mathcal{F}, \mathcal{A}, \delta^*, s_0, S_g \rangle$ be a ground-truth deterministic transition system specified in relational STRIPS.

- $\mathcal{F} = \{f_1, f_2, \ldots, f_n\}$ is a finite set of relational Boolean fluents. The state space is factored: $\mathcal{S} = 2^{\mathcal{F}}$, with cardinality $|\mathcal{S}| = 2^n$.
- $\mathcal{A}$ is a finite set of action schemas. Each action $a \in \mathcal{A}$ is defined by a triple:

$$a = \langle \mathrm{Pre}(a), \mathrm{Add}(a), \mathrm{Del}(a) \rangle \tag{1}$$

  where $\mathrm{Pre}(a) \subseteq \mathcal{F}$ represents preconditions, and $\mathrm{Add}(a), \mathrm{Del}(a) \subseteq \mathcal{F}$ denote disjoint add and delete effects ($\mathrm{Add}(a) \cap \mathrm{Del}(a) = \emptyset$).
- An action a is *applicable* in state $s \in \mathcal{S}$ if and only if $\mathrm{Pre}(a) \subseteq s$.
- The deterministic state transition function $\delta^* : \mathcal{S} \times \mathcal{A} \to \mathcal{S} \cup \{\bot\}$ is defined as:

$$\delta^*(s, a) = \begin{cases} (s \setminus \mathrm{Del}^*(a)) \cup \mathrm{Add}^*(a) & \text{if } \mathrm{Pre}^*(a) \subseteq s \\ \bot & \text{otherwise} \end{cases} \tag{2}$$

  where $\bot$ indicates execution failure (inapplicability).

A learned or candidate world model is denoted by $\widehat{M} = \langle \mathcal{F}, \mathcal{A}, \widehat{\delta}, s_0, S_g \rangle$, with candidate action schemas $\widehat{a} = \langle \widehat{\mathrm{Pre}}(a), \widehat{\mathrm{Add}}(a), \widehat{\mathrm{Del}}(a) \rangle$.

Definition 1 (Sparse Rule Intervention)**.** The syntactic distance between ground-truth model M^* and candidate model $\widehat{M}$ is defined by the Action-Language AST edit distance:

$$\Delta DSL(M^*, \widehat{M}) = \sum_{a \in \mathcal{A}} \Big(|\mathrm{Pre}^*(a) \triangle \widehat{\mathrm{Pre}}(a)| + |\mathrm{Add}^*(a) \triangle \widehat{\mathrm{Add}}(a)| + |\mathrm{Del}^*(a) \triangle \widehat{\mathrm{Del}}(a)| \Big) = k \tag{3}$$

When $k \ll |\mathcal{A}| \cdot |\mathcal{F}|$, the intervention is termed *sparse*. A *precondition omission* occurs when $\widehat{\mathrm{Pre}}(a) \subset \mathrm{Pre}^*(a)$ while effect sets are unmodified ($\widehat{\mathrm{Add}}(a) = \mathrm{Add}^*(a)$, $\widehat{\mathrm{Del}}(a) = \mathrm{Del}^*(a)$).

2 Formal Metric Definitions

2.1 State-Indexed Search Trees

Definition 2 (Search Tree)**.** For search horizon depth $D \in \mathbb{N}$, let $G_T = (V_T, E_T)$ denote the state-indexed search tree generated by exhaustive forward search from initial state s_0 under ground-truth dynamics M^*. Each node $u \in V_T$ corresponds to a reachable state-trajectory prefix $s(u) \in \mathcal{S}$. Each directed labeled edge $e = (u, a, v) \in E_T$ denotes an executable transition satisfying $\delta^*(s(u), a) = s(v) \neq \bot$. Similarly, let $\widehat{G}_T = (\widehat{V}_T, \widehat{E}_T)$ denote the search tree generated by candidate model $\widehat{M}$ from s_0 up to depth D.

2.2 Local Transition Prediction Accuracy ($A_{\mathbf{pred}}$)

Local evaluation assesses transition predictions over a finite dataset of transition queries $\mathcal{D}$.

Definition 3 (Local Accuracy)**.** Given a test dataset $\mathcal{D} = \{(s_i, a_i, s_i^*)\}_{i=1}^N$ where $s_i^* = \delta^*(s_i, a_i)$, the local prediction accuracy is:

$$A_{\mathrm{pred}}(\widehat{M}; \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{(s,a,s^*) \in \mathcal{D}} \mathbb{I}\left(\widehat{\delta}(s, a) = s^* \right) \tag{4}$$

We formalize two canonical test distributions:

1. **Reachable Valid Traces ($\mathcal{D}_{\text{valid}}$)**: Evaluated strictly along true environment transitions:

$$\mathcal{D}_{\text{valid}} = \{(s(u), a, s(v)) \mid (u, a, v) \in E_T(M^*)\} \tag{5}$$

2. **Search Fringe ($\mathcal{D}_{\text{fringe}}$)**: Evaluated across the union of physical transitions and candidate model expansions:

$$\mathcal{D}_{\text{fringe}} = E_T(M^*) \cup (\widehat{E}_T \setminus E_T(M^*)) \tag{6}$$

2.3 Global Execution Correctness: PESR and R_{play}

Let Planner$(\widehat{M}, s_0, g)$ denote an optimal cost-minimizing forward search planner (e.g., A^* with an admissible heuristic $h^{\text{LM-cut}}$) operating on candidate model $\widehat{M}$. The planner synthesizes a candidate plan $\pi^*_{\widehat{M}} = \langle a_1, a_2, \ldots, a_m \rangle$.

Definition 4 (Plan Execution Success Rate (PESR))**.** PESR measures whether the plan synthesized on model $\widehat{M}$ executes successfully in the physical environment M^* to reach goal $g \subseteq \mathcal{F}$:

$$\text{PESR}(\widehat{M}) = \begin{cases} 1 & \text{if } \delta^*(s_0, \pi^*_{\widehat{M}}) \models g \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

Definition 5 (Play Regret (R_{play}))**.** Let $C_{M^*}(\pi)$ denote the cumulative execution cost of plan π in the ground-truth environment M^*, defined as:

$$C_{M^*}(\pi) = \begin{cases} \sum_{t=1}^{|\pi|} \text{cost}(a_t) & \text{if } \forall t,\ \delta^*(s_{t-1}, a_t) \neq \perp \text{ and } \delta^*(s_0, \pi) \models g \\ \infty & \text{if } \exists t,\ \delta^*(s_{t-1}, a_t) = \perp \text{ (execution failure)} \end{cases} \tag{8}$$

The Play Regret of model $\widehat{M}$ relative to the ground-truth optimal plan $\pi^*_{M^*}$ is:

$$R_{\text{play}}(\widehat{M}) = C_{M^*}(\pi^*_{\widehat{M}}) - C_{M^*}(\pi^*_{M^*}) \tag{9}$$

Whenever an inexecutable plan is synthesized, $R_{\text{play}}(\widehat{M}) = \infty$.

2.4 Topological Search-Tree Gap Metric ($d_\triangle$)

Definition 6 (Topological Edit Distance)**.** The structural gap between the physical search tree G_T and the model search tree $\widehat{G}_T$ is defined as the symmetric difference of their edge sets:

$$d_\triangle(G_T, \widehat{G}_T) = |E_T \triangle \widehat{E}_T| = |E_T \setminus \widehat{E}_T| + |\widehat{E}_T \setminus E_T| \tag{10}$$

Edges in $E_{\text{phantom}} = \widehat{E}_T \setminus E_T$ are termed *phantom edges.*

3 Formal Proofs

3.1 Lemma 1: Monotone Edge Addition under Precondition Omission

Lemma 7 (No-Edges-Lost Lemma)**.** *Let $M^* = \langle \mathcal{F}, \mathcal{A}, \delta^*, s_0, S_g \rangle$ be the ground-truth transition system. Suppose $\widehat{M}$ is constructed such that for all actions $a \in \mathcal{A}$:*

$$\widehat{Pre}(a) \subseteq Pre^*(a), \quad \widehat{Add}(a) = Add^*(a), \quad \widehat{Del}(a) = Del^*(a) \tag{11}$$

Then every edge in the true search tree G_T is preserved in candidate search tree $\widehat{G}_T$:

$$E_T(M^*) \subseteq \widehat{E}_T \quad \textit{and} \quad |E_T \setminus \widehat{E}_T| = 0 \tag{12}$$

Consequently, the topological gap simplifies strictly to the number of phantom edges:

$$d_\triangle(G_T, \widehat{G}_T) = |\widehat{E}_T \setminus E_T| \geq 0 \tag{13}$$

Proof. Let $e = (u, a, v) \in E_T(M^*)$ be an arbitrary edge in the ground-truth search tree G_T. By definition of G_T, this edge corresponds to a valid physical transition:

$$\delta^*(s(u), a) = s(v) \neq \bot \tag{14}$$

This implies that action a is applicable at state $s(u)$ in M^*:

$$\mathrm{Pre}^*(a) \subseteq s(u) \tag{15}$$

By hypothesis, the candidate model satisfies $\widehat{\mathrm{Pre}}(a) \subseteq \mathrm{Pre}^*(a)$. By transitivity of subset inclusion:

$$\widehat{\mathrm{Pre}}(a) \subseteq s(u) \tag{16}$$

Hence, action a is applicable at state $s(u)$ in $\widehat{M}$. Next, compute the successor state $\widehat{\delta}(s(u), a)$:

$$\widehat{\delta}(s(u), a) = (s(u) \setminus \widehat{\mathrm{Del}}(a)) \cup \widehat{\mathrm{Add}}(a) \tag{17}$$

Because effect sets are unmodified ($\widehat{\mathrm{Del}}(a) = \mathrm{Del}^*(a)$ and $\widehat{\mathrm{Add}}(a) = \mathrm{Add}^*(a)$):

$$\widehat{\delta}(s(u), a) = (s(u) \setminus \mathrm{Del}^*(a)) \cup \mathrm{Add}^*(a) = \delta^*(s(u), a) = s(v) \tag{18}$$

We now proceed by induction on search tree depth $t \in [0, D]$.

- **Base Case ($t = 0$)**: At the root node u_0, $s(u_0) = s_0 \in V_T$ and $s_0 \in \widehat{V}_T$.
- **Inductive Step**: Assume that for all nodes u at depth $t < D$, $u \in V_T \implies u \in \widehat{V}_T$. For any edge $(u, a, v) \in E_T$, action a is applicable in $\widehat{M}$ and produces identical successor $s(v)$. Thus node $v \in \widehat{V}_T$ and edge $(u, a, v) \in \widehat{E}_T$.

Therefore, $E_T \subseteq \widehat{E}_T$, which implies $E_T \setminus \widehat{E}_T = \emptyset$, so $|E_T \setminus \widehat{E}_T| = 0$. Substituting into the definition of symmetric difference:

$$d_\triangle(G_T, \widehat{G}_T) = |E_T \setminus \widehat{E}_T| + |\widehat{E}_T \setminus E_T| = 0 + |\widehat{E}_T \setminus E_T| = |\widehat{E}_T \setminus E_T| \tag{19}$$

This completes the proof. □

3.2 Proposition 1: Search-Tree Phantom Path Divergence

Proposition 8 (Topological Divergence Bound)**.** *Let G_T be a regular b-ary search tree of uniform branching factor $b \geq 2$ and depth horizon $D \in \mathbb{N}$. Suppose candidate model $\widehat{M}$ is constructed by omitting a single bottleneck precondition $p^* \in Pre^*(a^*)$ for an action a^* applicable at depth $d < D$. Under the Fresh-Subtree Assumption (the unblocked state expands a regular tree of depth $D - (d + 1)$ with branching factor b), the topological gap and local prediction accuracy satisfy:*

$$d_\triangle(G_T, \widehat{G}_T) = \Omega\left(b^{D-d}\right) \tag{20}$$

and evaluated over the candidate search distribution $\mathcal{D}_{fringe} = E_T \cup (\widehat{E}_T \setminus E_T)$:

$$A_{pred}(\widehat{M}; \mathcal{D}_{fringe}) \geq 1 - b^{-d} \tag{21}$$

Furthermore, for any precision threshold $\epsilon > 0$, whenever the bottleneck depth satisfies $d \geq \lceil \log_b(1/\epsilon) \rceil$:

$$A_{pred}(\widehat{M}) \geq 1 - \epsilon \quad \textit{while} \quad d_\triangle(G_T, \widehat{G}_T) \xrightarrow{D \to \infty} \infty \tag{22}$$

Proof. We count the cardinality of the respective edge sets. In a regular b-ary search tree of uniform depth D, the total number of physical edges in G_T is:

$$|E_T| = \sum_{i=0}^{D-1} b^{i+1} = b \cdot \frac{b^D - 1}{b-1} = \Theta(b^D) \tag{23}$$

At depth $d \in [0, D-1]$, omitting precondition p^* unblocks action a^* at a state s_d, creating an initial phantom edge $(s_d, a^*, s_{\text{phantom}})$. By the Fresh-Subtree Assumption, the node s_{phantom} is expanded recursively to horizon D, forming a complete subtree of depth $D-(d+1)$. The number of phantom edges generated in this unblocked subtree is:

$$|E_{\text{phantom}}| = 1 + \sum_{j=1}^{D-d-1} b^j = 1 + b \cdot \frac{b^{D-d-1} - 1}{b-1} = \frac{b^{D-d} - 1}{b-1} = \Theta\left(b^{D-d}\right) \tag{24}$$

By Theorem 7, $E_T \subseteq \widehat{E}_T$, so the topological gap is:

$$d_{\triangle}(G_T, \widehat{G}_T) = |E_{\text{phantom}}| = \frac{b^{D-d} - 1}{b-1} = \Omega\left(b^{D-d}\right) \tag{25}$$

Now consider local prediction accuracy over the candidate search distribution:

$$\mathcal{D}_{\text{fringe}} = E_T \cup E_{\text{phantom}}, \quad |\mathcal{D}_{\text{fringe}}| = |E_T| + |E_{\text{phantom}}| \tag{26}$$

By Theorem 7, for every edge in E_T, model $\widehat{M}$ predicts the exact ground-truth transition. The model errs exclusively on the phantom transitions in E_{phantom}, where $\widehat{M}$ predicts a valid state change while ground truth M^* returns $\perp$. Thus:

$$A_{\text{pred}}(\widehat{M}; \mathcal{D}_{\text{fringe}}) = \frac{|E_T|}{|E_T| + |E_{\text{phantom}}|} \tag{27}$$

Substituting the exact sums:

$$A_{\text{pred}}(\widehat{M}) = \frac{b\frac{b^D-1}{b-1}}{b\frac{b^D-1}{b-1} + \frac{b^{D-d}-1}{b-1}} = \frac{b(b^D-1)}{b(b^D-1) + (b^{D-d}-1)} \tag{28}$$

$$= \frac{1}{1 + \frac{b^{D-d}-1}{b(b^D-1)}} = \frac{1}{1 + b^{-d}\left(\frac{1-b^{-(D-d)}}{b(1-b^{-D})}\right)} \tag{29}$$

Since $b \geq 2$ and $D > d \geq 1$, we have:

$$\frac{1 - b^{-(D-d)}}{b(1-b^{-D})} \leq \frac{1}{b} < 1 \tag{30}$$

Using the inequality $\frac{1}{1+x} \geq 1 - x$ for $x \geq 0$:

$$A_{\text{pred}}(\widehat{M}; \mathcal{D}_{\text{fringe}}) \geq 1 - b^{-d} \tag{31}$$

Finally, choose any $\epsilon > 0$. Setting $d \geq \lceil \log_b(1/\epsilon) \rceil$ implies $b^{-d} \leq \epsilon$, guaranteeing:

$$A_{\text{pred}}(\widehat{M}; \mathcal{D}_{\text{fringe}}) \geq 1 - \epsilon \tag{32}$$

Simultaneously, holding d fixed and taking the horizon limit $D \to \infty$:

$$\lim_{D\to\infty} d_{\triangle}(G_T, \widehat{G}_T) = \lim_{D\to\infty} \frac{b^{D-d} - 1}{b-1} = \infty \tag{33}$$

This concludes the proof. □

3.3 Proposition 2: Causal Decoupling and Statistical Independence of A_{pred} and PESR

Proposition 9 (Causal Decoupling Theorem)**.** *Local prediction accuracy A_{pred} and global plan execution success rate PESR are functionally and statistically decoupled. Specifically:*

1. ***High Accuracy with Total Execution Collapse**: For any arbitrary tolerance $\epsilon > 0$, there exists an infinite family of planning tasks Π and candidate models $\widehat{M}$ such that:*

$$A_{pred}(\widehat{M}) \geq 1 - \epsilon \quad \textit{and} \quad PESR(\widehat{M}) = 0 \quad (R_{play}(\widehat{M}) = \infty) \tag{34}$$

2. ***Low Accuracy with Perfect Optimal Execution**: Conversely, for any $\epsilon > 0$, there exist models $\widehat{M}'$ exhibiting arbitrarily low local accuracy:*

$$A_{pred}(\widehat{M}') \leq \epsilon \quad \textit{and} \quad PESR(\widehat{M}') = 1 \quad (R_{play}(\widehat{M}') = 0) \tag{35}$$

Consequently, there exists no non-trivial monotonic lower-bounding function $f : [0, 1] \to [0, 1]$ satisfying $PESR(\widehat{M}) \geq f(A_{pred}(\widehat{M}))$ with $f(1 - \epsilon) > 0$.

Proof. We prove both parts constructively.

Part 1: Constructing $A_{\text{pred}} \geq 1 - \epsilon$ with PESR $= 0$. Construct a parameterized planning task Π_d with horizon $D \gg d$. Let the physical environment M^* possess an optimal execution path $\pi^* = \langle a_1^*, a_2^*, \ldots, a_L^* \rangle$ from s_0 to goal g with path length $L = D$ and uniform cost $C(\pi^*) = L$. At search depth $d = \lceil \log_b(1/\epsilon) \rceil$, introduce an action a_{shortcut} that is physically blocked by bottleneck precondition $p^* \in \text{Pre}^*(a_{\text{shortcut}})$ (i.e., $s_d \not\models p^*$). If a_{shortcut} were executable, its transition reaches goal state g directly with action cost 1. Thus, any path utilizing a_{shortcut} has total planned cost:

$$C(\pi_{\text{phantom}}) = d + 1 < L \tag{36}$$

Now, let candidate model $\widehat{M}$ be constructed by omitting p^* from a_{shortcut} ($\widehat{\text{Pre}}(a_{\text{shortcut}}) = \text{Pre}^*(a_{\text{shortcut}}) \setminus \{p^*\}$), while all other actions and effects are perfectly specified.

- By Theorem 8, the local accuracy of $\widehat{M}$ satisfies:

$$A_{\text{pred}}(\widehat{M}; \mathcal{D}_{\text{fringe}}) \geq 1 - b^{-d} \geq 1 - \epsilon \tag{37}$$

- We now evaluate the cost-minimizing planner $\text{Planner}(\widehat{M}, s_0, g)$. Because the planner searches for minimal-cost paths, it evaluates:

$$\text{cost}(\pi_{\text{phantom}}) = d + 1 < L = \text{cost}(\pi^*) \tag{38}$$

  Therefore, the planner deterministically selects plan $\pi^*_{\widehat{M}} = \pi_{\text{phantom}} = \langle a_1^*, \ldots, a_d^*, a_{\text{shortcut}} \rangle$.

- When $\pi^*_{\widehat{M}}$ is executed in the physical environment M^*, prefix $\langle a_1^*, \ldots, a_d^* \rangle$ executes successfully to reach state s_d. At step $d + 1$, action a_{shortcut} is executed. Since $s_d \not\models p^*$, we have:

$$\delta^*(s_d, a_{\text{shortcut}}) = \bot \tag{39}$$

  The plan halts catastrophically before reaching goal g. Consequently, $\text{PESR}(\widehat{M}) = 0$ and $R_{\text{play}}(\widehat{M}) = \infty$.

Part 2: Constructing $A_{\text{pred}} \leq \epsilon$ with PESR $= 1$. Let $\pi^* = \langle a_1^*, a_2^*, \ldots, a_m^* \rangle$ be the ground-truth optimal plan for task Π, traversing states $\langle s_0, s_1^*, \ldots, s_m^* \rangle$ with $s_m^* \models g$. Augment the environment with a set of K distractor actions $\mathcal{A}_{\text{distractor}}$ that govern task-irrelevant fluents $\mathcal{F}_{\text{distractor}}$, where $|\mathcal{A}_{\text{distractor}}| \geq \frac{m}{\epsilon}$. Construct model $\widehat{M}'$ such that:

1. $\widehat{M}'$ specifies the actions along the optimal path $\{a_1^*, \ldots, a_m^*\}$ with 100% fidelity.
2. For all distractor actions $a \in \mathcal{A}_{\text{distractor}}$, $\widehat{M}'$ predicts completely incorrect transitions (e.g., predicting $\perp$ for valid transitions or invalid state updates).

Evaluating local accuracy over the full transition distribution yields:

$$A_{\text{pred}}(\widehat{M}') \leq \frac{m}{m + |\mathcal{A}_{\text{distractor}}|} \leq \frac{m}{m + m/\epsilon} \leq \epsilon \tag{40}$$

However, when the forward planner operates on $\widehat{M}'$, all distractor actions are either perceived as inexecutable or leading to unpromising states. The planner deterministically outputs the unique valid path π^*. Executing π^* in M^* succeeds with zero regret:

$$\text{PESR}(\widehat{M}') = 1 \quad \text{and} \quad R_{\text{play}}(\widehat{M}') = C_{M^*}(\pi^*) - C_{M^*}(\pi^*) = 0 \tag{41}$$

Non-Existence of Monotone Bound. Suppose for contradiction there exists a function $f : [0, 1] \to [0, 1]$ such that $\text{PESR}(\widehat{M}) \geq f(A_{\text{pred}}(\widehat{M}))$ for all $\widehat{M}$, with $f(1 - \epsilon) > 0$. Applying this bound to Part 1 implies:

$$0 = \text{PESR}(\widehat{M}) \geq f(A_{\text{pred}}(\widehat{M})) \geq f(1 - \epsilon) > 0 \tag{42}$$

which is an immediate contradiction. Hence, no such lower bound exists, establishing that local prediction accuracy and global plan execution are functionally decoupled. □

4 Conclusion

The formal proofs established in Theorem 7, Theorem 8, and Theorem 9 provide the theoretical foundation for this thesis. They prove that the verified-vs-correct gap in symbolic world models is not an empirical artifact or a quirk of specific training runs, but a fundamental property of combinatorial search over state-space graphs. Evaluating world models via passive transition accuracy is mathematically incapable of certifying autonomous agents for goal-directed play. Resolving this challenge necessitates active, counterexample-guided repair mechanisms such as CEG-OMR.